



The Bombay Salesian Society's
DON BOSCO INSTITUTE OF TECHNOLOGY

An Autonomous Institute Affiliated to University of Mumbai
Premier Automobiles Road, Kurla West, Mumbai - 400070



Radar and Automotive Radar Systems session
Report on - Radar and Automotive Radar

Title: Radar and Automotive Radar

Date: 11th April, 2026

Time: 12:30 p.m. – 1:30 p.m.

Venue: Online (zoom)

Target Audience: Open For All

No. of Participants Present: 35

No. of Girl Participants Present: 9

No. of Boy Participants Present: 26

Resource Person: Mr Jason Fernandes

Organization of Resource Person: BMW AG

Organizing Department: IEEE-DBIT

Faculty Coordinator: Ms Freda Carvalho

Objectives:

- To introduce students to the fundamentals of radar systems and their real-world applications
- To provide understanding of automotive radar technologies (FMCW, 76–81 GHz)
- To expose students to advanced concepts like phased array & beam steering
- To highlight the role of AI in modern radar systems
- To create awareness about career and research opportunities in RF & Radar Engineering

Outcomes:

- Students gained knowledge of how radar measures range, velocity, and direction
- Students understood working of FMCW radar used in automotive systems
- Students explored latest technologies like MIMO, AI-based detection, and sensing
- Students became aware of current research areas and industry trends
- Students developed interest in core electronics fields like RF, radar, and embedded systems



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Detailed Report:

A guest lecture titled “*Introduction to Radar Systems: Enabling the Future of Autonomous Mobility*” was organized by the IEEE MTT-S (Microwave Theory & Technology Society) Student Chapter in association with IEEE-DBIT, under the *Technology for the Young* initiative. The event was conducted on 11th April 2026 from 12:30 PM to 1:30 PM via Zoom and was free and open to all students. The session was delivered by Mr. Jason Fernandes, Radar Expert at BMW AG, who provided valuable industry insights and practical knowledge to the participants.

The lecture was primarily aimed at students from the Electronics and Telecommunication (EXTC) branch, along with others interested in RF, microwave, and radar engineering. The session began with an introduction to the basic principles of radar systems, explaining how electromagnetic waves are used to detect objects and measure parameters such as range, velocity, and direction. A major focus was on automotive radar technologies, particularly Frequency-Modulated Continuous Wave (FMCW) radar operating in the 76–81 GHz frequency band, widely used in modern vehicles for applications like adaptive cruise control and collision avoidance.

The speaker further discussed advanced concepts such as phased array antennas and beam steering, which enable directional control of radar signals, as well as MIMO (Multiple Input Multiple Output) systems that enhance detection accuracy and resolution. The session also highlighted the role of high-resolution radar chips and the growing importance of Artificial Intelligence in improving object detection and sensing capabilities. Emerging technologies such as 4D radar sensing were introduced, providing students with insights into the future of autonomous mobility.

In addition to technical concepts, the lecture shed light on current research trends and career opportunities in RF and radar engineering. Students gained an understanding of the interdisciplinary nature of the field, involving signal processing, communication systems, and embedded technologies. The speaker also emphasized the skills required to pursue careers in industries such as automotive, defense, and wireless communication.

Overall, the session was highly informative and successfully met its objectives. Students developed a clear understanding of radar fundamentals and automotive applications, while also gaining exposure to advanced technologies like AI-based detection and 4D sensing. The lecture sparked interest in core electronics domains and created awareness about the wide range of research and career opportunities in radar and autonomous systems.

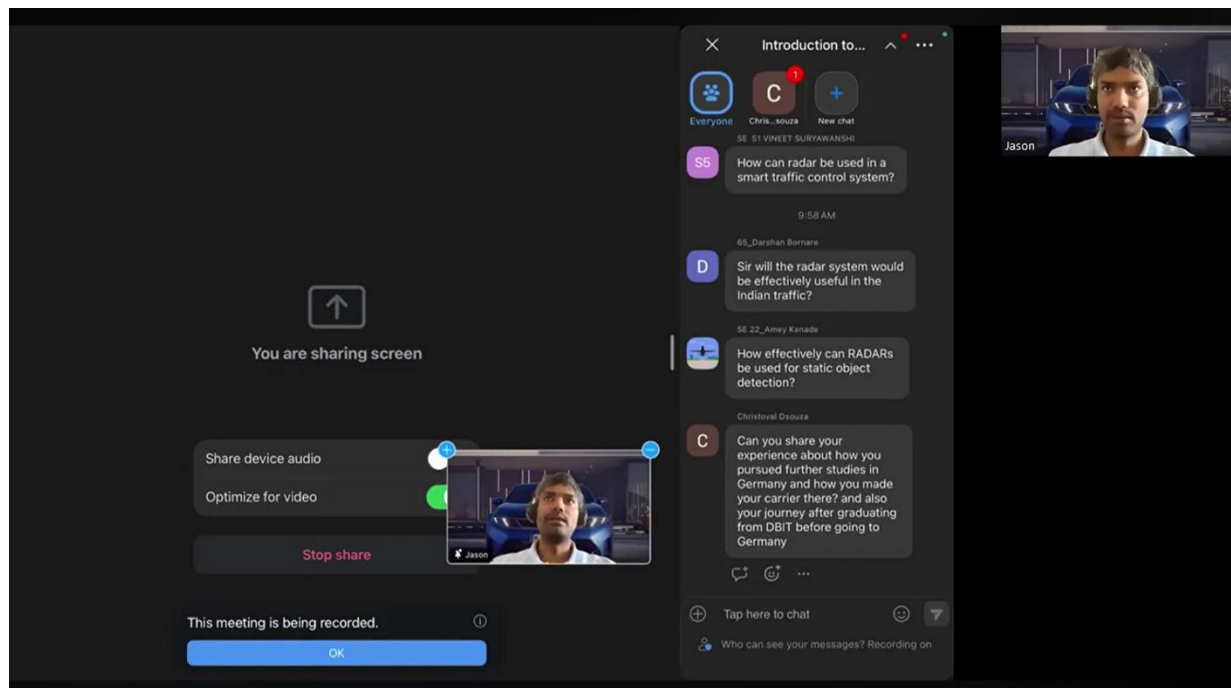


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Snapshot of the Event:





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Automotive Radar



Sensing the road in 4D

Jason Fernandes. Radar Expert BMW AG



Samiksha Gaonkar



Operation principle

- FMCW Modulation in 76 - 81 GHz
- Phased array based electronic beam steering
- AI based object recognition
- AI based segmentation and road feature recognition



C

Christoval Drouza

J

Jason



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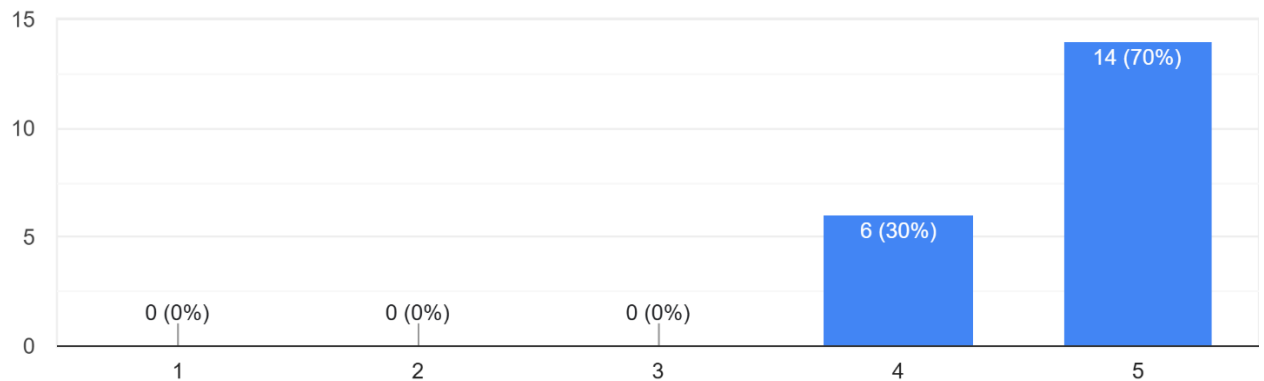
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Feedback Analysis:

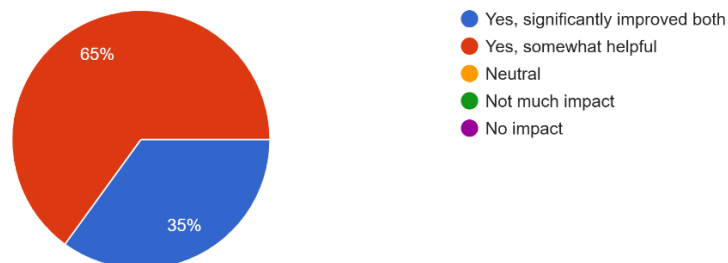
How would you rate your Overall Experience

20 responses



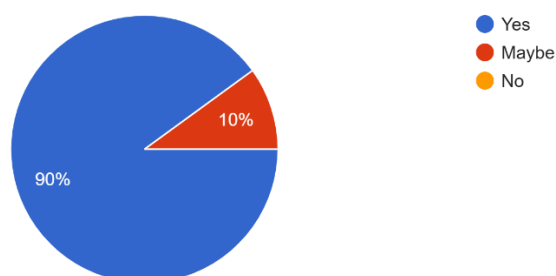
Did the session increase your interest in RF/Radar Engineering and benefit your learning/career?

20 responses



Would you like more such industry sessions?

20 responses





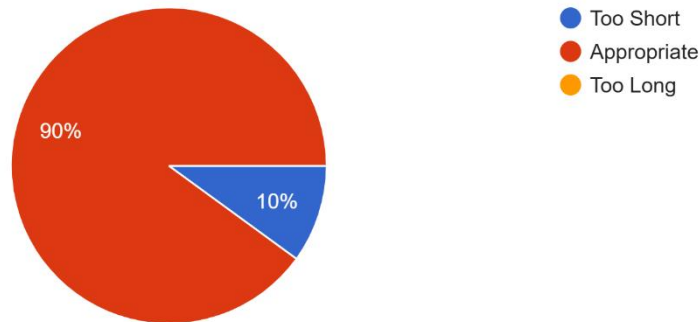
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Was the duration of the session appropriate?

20 responses



The feedback collected from participants of the “Introduction to Radar & Automotive Radar Systems” session indicates an overwhelmingly positive response, reflecting the effectiveness and relevance of the lecture. A total of 20 responses were recorded, with 70% of participants rating their overall experience as excellent (5/5) and the remaining 30% rating it as very good (4/5), demonstrating a high level of satisfaction. Furthermore, all respondents acknowledged that the session contributed positively to their learning and career awareness, with 35% stating it significantly enhanced both their interest and understanding in RF/Radar Engineering, while 65% found it somewhat beneficial. The demand for similar industry-oriented sessions is notably high, with 90% expressing a strong interest in attending more such events and the remaining participants indicating a possible interest. In terms of session duration, 90% felt that the timing was appropriate, whereas a small fraction suggested extending it for deeper exploration.

Qualitative feedback highlights several constructive suggestions and emerging interests among students. Many participants appreciated the session content but expressed a desire for more **interactive and hands-on learning experiences**, such as workshops and practical demonstrations. There is also a noticeable inclination toward expanding topics into adjacent domains like **VLSI design, embedded systems, DSP, communication systems, and processor manufacturing**. Some responses specifically emphasized the importance of **industry-relevant tools (e.g., MATLAB, Cadence)** and deeper exposure to **mmWave radar, RF front-end circuits, and real-world implementations**. Additionally, students showed interest in **career-oriented sessions, alumni talks, and insights into internships and research opportunities**, indicating a strong focus on practical career development. A few participants also suggested conducting offline or hybrid workshops to enhance engagement. Emerging technology trends such as **AI, edge computing, quantum technologies, robotics, and sustainable engineering** were also mentioned as potential areas for future events.

Overall, the feedback clearly demonstrates that the session successfully achieved its objectives of increasing awareness, interest, and understanding of radar systems, while also highlighting opportunities to further enhance future IEEE events through increased interactivity, practical exposure, and broader technological coverage.



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Event Poster:

  **MTT-S**
IEEE MICROWAVE THEORY & TECHNOLOGY SOCIETY

presents

INTRODUCTION TO RADAR SYSTEMS

Enabling the Future of
Autonomous Mobility

**11TH
APRIL**
12:30 PM – 01:30 PM

WHAT YOU WILL GAIN

- ▶ FMCW Radar & 4D Sensing Fundamentals
- ▶ Phased Array Beamforming, MIMO & High-Resolution Chips
- ▶ AI in Automotive Radar Systems & Emerging Research Areas

 **Free and Open to all !**

JOIN NOW! 

Scan to Register



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 **CONTACT:** Samiksha Gaonkar - 7021814110
Janhvi Shejwal - 9421974916



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Social Media Links:

https://www.instagram.com/ieee_dbit/

Registration Details:

S. No.	Name	Branch
1.	Samiksha Gaonkar	EXTC
2.	Christoval Dsouza	EXTC
3.	Janhavi Shejwal	EXTC
4.	Mohd Raza	EXTC
5.	Amey Kanade	EXTC
6.	Kunal Jadhav	EXTC
7.	Aarya Jadhav	EXTC
8.	Rohit Kumar	EXTC
9.	Jordan Pinto	EXTC
10.	Ashlesha Tambat	EXTC
11.	Yash Khanvilkar	EXTC
12.	Abhinav Borse	EXTC
13.	Raj Vats	EXTC
14.	Disha Mishra	EXTC
15.	Aditya Jadhav	EXTC
16.	Neha Haldankar	EXTC
17.	Tirth Shetty	EXTC
18.	Amaan Waghani	EXTC
19.	Darshan Bornare	EXTC
20.	Harshit Patil	EXTC
21.	Raza Sayed	EXTC
22.	Amit P	EXTC
23.	Yusra Ansari	EXTC
24.	Rishi Bajpai	EXTC
25.	Swara Ghosalkar	EXTC
26.	Vineet Suryawanshi	EXTC
27.	Dhruv Tare	EXTC
28.	Vaibhav Kumar Jha	EXTC
29.	Deviprasad Shetty	IT
30.	Sanket Jadhav	COMPS
31.	Tanmay Tajane	IT
32.	Prem Singh	EXTC
33.	Aayush Borle	EXTC
34.	Kenneth Dsouza	EXTC
35.	Tanvi Kamitkar	EXTC



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Report Prepared By: IEEE-DBIT

Name of the Student: Aarya Jadhav &
Shreya Manoj

Post of the student:
Reporting Head & Joint Reporting Head,
IEEE-DBIT

Report Approved By: IEEE - DBIT

Name of the Faculty: Ms. Freda
Carvalho

Post of the Faculty: IEEE-DBIT
Student Branch Counsellor